P P SAVANI UNIVERSITY

SCHOOL OF ENGINEERING

B.TECH FINAL PROJECT REPORT

(Hard-Bound Project Report Format)

PARKSMART: REAL-TIME SMART PARKING RESERVATION & AUTOMATED SECURITY GATE ACCESS MANAGEMENT SYSTEM

Particular	Details
Project Title	ParkSmart: Real-Time Smart Parking Reservation & Automated Security Gate Access Management System
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Enrollment Number	**23SE02CE064**
Program / Branch	B.Tech - Computer Engineering
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Company / Organization	P P Savani University (University UDP Project)
Department / Domain	Full-Stack Web Development & Smart Transportation Systems
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Academic Year	2026–2027

P P SAVANI UNIVERSITY | SCHOOL OF ENGINEERING

CERTIFICATE OF COMPLETION

This is to certify that **KANANI MYUSI UDAYBHAI**, Enrollment No.: **23SE02CE064**, student of **B.Tech Computer Engineering, 8th Semester**, has successfully completed the Final Year Project titled:

"PARKSMART: REAL-TIME SMART PARKING RESERVATION & AUTOMATED SECURITY GATE ACCESS MANAGEMENT SYSTEM"

during the academic term at **P P Savani University** under the guidance and supervision of Institute Mentors.

Date of Issue: 12 / 10 / 2026

_____
Mentor (Company / Industry)

_____
Mentor (Institute / Internal)

DECLARATION

I hereby declare that this project report entitled **"ParkSmart: Real-Time Smart Parking Reservation & Automated Security Gate Access Management System"** is an authentic record of our engineering work carried out during the academic year 2026–2027.

The work embodied in this project has not been submitted elsewhere for the award of any other degree or diploma. All external references, libraries, and frameworks have been duly acknowledged and cited according to academic standards.

KANANI MYUSI UDAYBHAI
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CHAPTER 4: REQUIREMENTS ANALYSIS & PROJECT SCOPE

4.1 Functional Requirements Specification

The functional requirements outline the core operations, security authentications, dynamic pass generation, and automated gate clearance processes designed and implemented in the ParkSmart system:

Table 4.1: Functional Requirements Specification

Req. ID	Requirement	Priority	Source / Basis	Status
FR-01	User registration, login, and JWT session authorization	High	System Security	Completed
FR-02	Real-time visual 2D parking bay matrix	High	User Experience	Completed
FR-03	Slot reservation with vehicle type & duration filters	High	Core Function	Completed
FR-04	HMAC-SHA256 Dynamic QR token generation (30 s rotation)	High	Anti-Fraud	Completed
FR-05	Security Desk camera scanner with Smart Auto Mode	High	Gate Operations	Completed
FR-06	1-Click Gate mode recovery for active/confirmed vehicles	Medium	Error Prevention	Completed
FR-07	Automated overstay penalty billing at checkout	High	Billing Module	Completed
FR-08	Admin dashboard for slot management and revenue logs	High	Management	Completed

4.2 Non-Functional Requirements Specification

The non-functional specifications define the system's operational standards, security measures, responsiveness across mobile and desktop devices, and real-time reliability:

Table 4.2: Non-Functional Requirements Specification

Req. ID	Category	Requirement	Target / Measure	Status
NFR-01	Performance	API response time < 200 ms under standard loads	< 200 ms	Met
NFR-02	Security	bcrypt password hashing (10 salt rounds), JWT expiration	10 salt rounds	Met

NFR-03	Responsiveness	Fluid layout across 320 px to 1920 px viewports	320 px–1920 px	Met
NFR-04	Reliability	±15-minute clock drift tolerance for dynamic tokens	±15 minutes	Met
NFR-05	Availability	Zero-configuration camera fallback on web browsers	Multi-camera auto fallback	Met

4.3 Hardware Requirements

The hardware configuration covers client access devices, security desk optical input hardware, cloud hosting compute nodes, and simulated barrier equipment:

Table 4.3: Hardware Requirements

Component	Specification	Purpose	Availability
Client Device	Chrome, Edge, Firefox or Safari on Windows, Android, macOS or iOS	Driver, admin and security interfaces	Available
Camera	Standard webcam, smartphone rear/front camera or USB camera	QR scanning at security gate desk	Available
Server	Node.js v18+, MongoDB v6.0+, minimum 2 GB RAM	Hosting REST API and database	Available
Network	Active internet connection	Real-time verification against cloud database	Available
Boom Barrier Controller	ESP32 / Arduino with servo (future scope)	Physical gate control	Simulated via UI

4.4 Software Requirements Specification

The production software stack utilizes open-source enterprise libraries and cloud-native managed solutions:

Table 4.4: Software Requirements Specification

Software / Tool	Version	Purpose	Source / License
Node.js	v20+	Backend runtime	Open source (MIT)
MongoDB Atlas	6.0+	Cloud NoSQL database	Free tier / SSPL
React	18.2	Frontend UI framework	Open source (MIT)
Tailwind CSS	3.4	Responsive styling framework	Open source (MIT)
TypeScript	ES2022 target	Type-safe full-stack development	Open source (Apache-2.0)
Express.js	4.18	REST routing and middleware	Open source (MIT)
Html5Qrcode	2.3+	In-browser QR scanning	Open source (Apache-2.0)
qrcode.react	Stable release	QR code rendering for driver pass	Open source (ISC)

bcryptjs / jsonwebtoken	Stable release	Password hashing and JWT sessions	Open source (MIT)
Postman / Git & GitHub	Latest	API testing and version control	Free tier

4.5 Data / Dataset Requirements

The application utilizes schema-enforced, transactional NoSQL collections generated at runtime during system operations:

Table 4.5: Data / Dataset Requirements

Data Source	Type	Size / Records	Pre-processing	Purpose
Application-generated data	Users collection	Created at runtime (user registrations)	Password hashing (bcrypt), validation	Authentication and RBAC
Application-generated data	Slots collection	Parking bays configured by admin	Normalization of slot type and status	Floor grid and availability
Application-generated data	Bookings collection	Created per reservation	Time-window validation, QR token generation	Reservations and gate validation
Application-generated data	SecurityLogs & Blacklists	One record per gate scan / blocked vehicle	Timestamp, gate mode tagging, uppercase vehicle numbers	Audit and access control

4.6 Constraints and Assumptions

- **Internet dependency:** Verification of dynamic QR tokens is performed against the cloud database and requires continuous network connectivity.
- **Browser camera access:** Video streams require a modern browser and explicit user media permission (HTTPS or localhost).
- **Simulated hardware / test-mode payment:** The physical boom barrier is simulated through software triggers and Razorpay transactions operate in test mode.
- **Clock synchronization assumption:** The client smartphone clock and server clock differ by no more than ±15 minutes (handled by ±30 token steps drift window).
- **Operational scope assumption:** The facility operates as a centralized multi-floor parking lot managed by a unified administrative team.

CHAPTER 5: METHODOLOGY & PLANNING

5.1 Agile Methodology

The project followed an iterative Agile Scrum methodology across four distinct development sprints: Sprint 1 focused on architecture, Mongoose schemas, and JWT auth; Sprint 2 delivered the visual 2D floor grid; Sprint 3 added the dynamic HMAC QR and camera scanner; Sprint 4 completed cloud deployment on Vercel and load testing.

5.2 Overall Project Workflow

The end-to-end user and vehicle journey is illustrated below. All operations—from account authentication and slot selection to cryptographic checkout, gate verification, and automatic overstay billing—are fully connected:

Figure 5.1: Overall Project Workflow (Scalable Vector Flowchart)

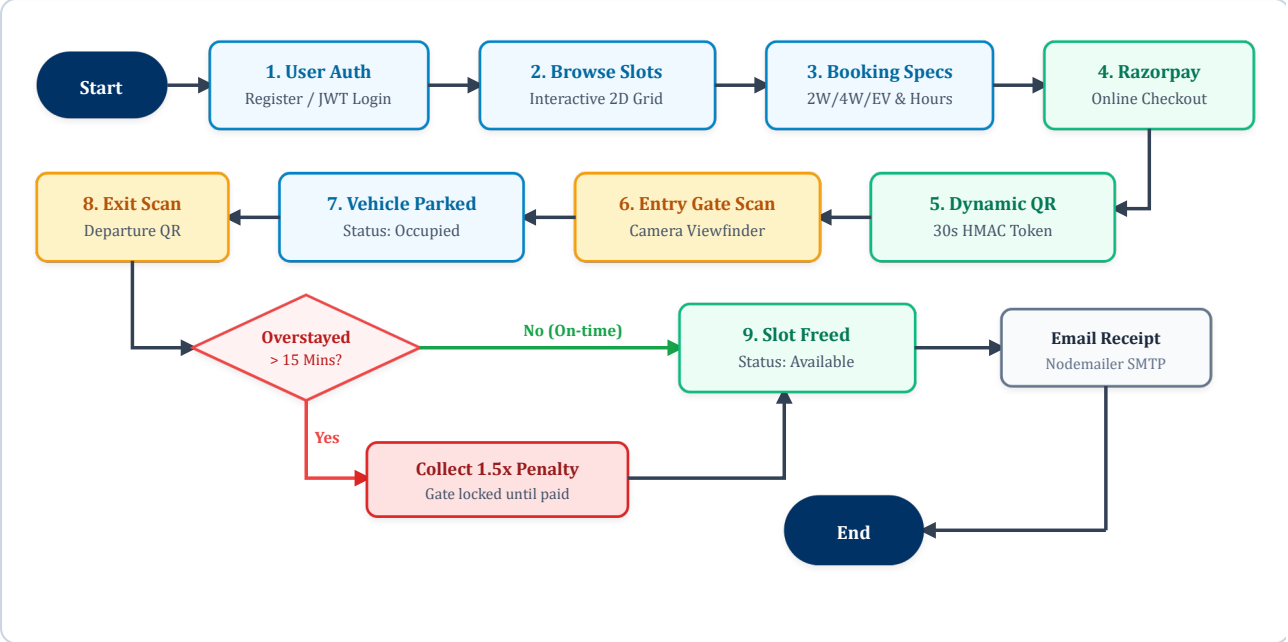


Figure 5.1: High-resolution end-to-end lifecycle flowchart of ParkSmart with automatic overstay evaluation.

CHAPTER 6: SYSTEM ANALYSIS & DESIGN

6.1 3-Tier Cloud Architecture

ParkSmart is deployed on a modern, decoupled cloud architecture separating the client presentation layer, serverless Node.js backend logic, and cloud database tier:

Figure 6.1: Deployed System Architecture of ParkSmart (3-Tier Cloud)

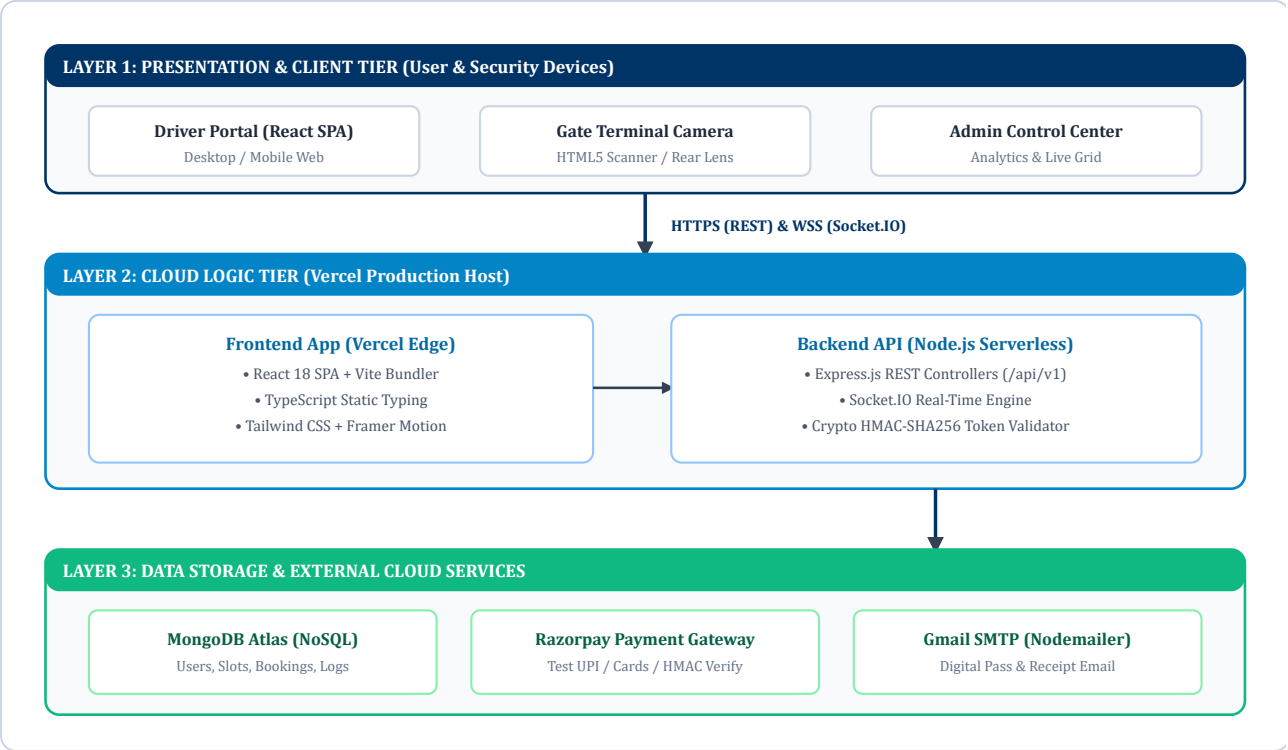


Figure 6.1: Cloud architecture separating Vercel frontend, Vercel serverless backend, MongoDB Atlas, and Razorpay SDK.

6.2 Security Staff Smart Auto Gate Controller

To ensure fast vehicle processing under 1.5 seconds, the camera scanner dynamically detects entry versus exit without requiring the guard to toggle modes manually:

Figure 6.2.3: Security Staff Smart Auto Gate Process Flow

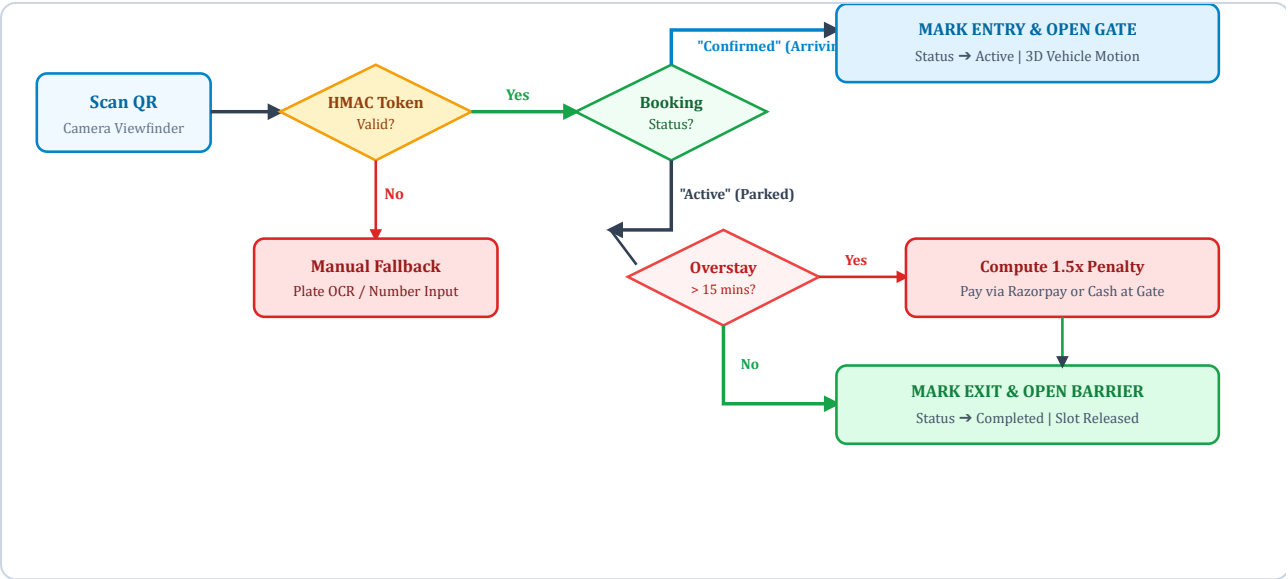


Figure 6.2.3: Gate access decision logic handling dynamic token validation, automatic routing, and penalty checkout.

6.3 Dynamic QR Pass Generation & Verification Workflow

To ensure cryptographic integrity and prevent ticket duplication or screen-capture sharing, ParkSmart generates a time-stepped HMAC-SHA256 digital token rotating every 30 seconds. The sequence diagram below illustrates the complete 10-step authentication, clock-drift validation, and automated boom barrier trigger:

Figure 6.3.5: Dynamic QR Pass Generation & Verification Sequence Diagram

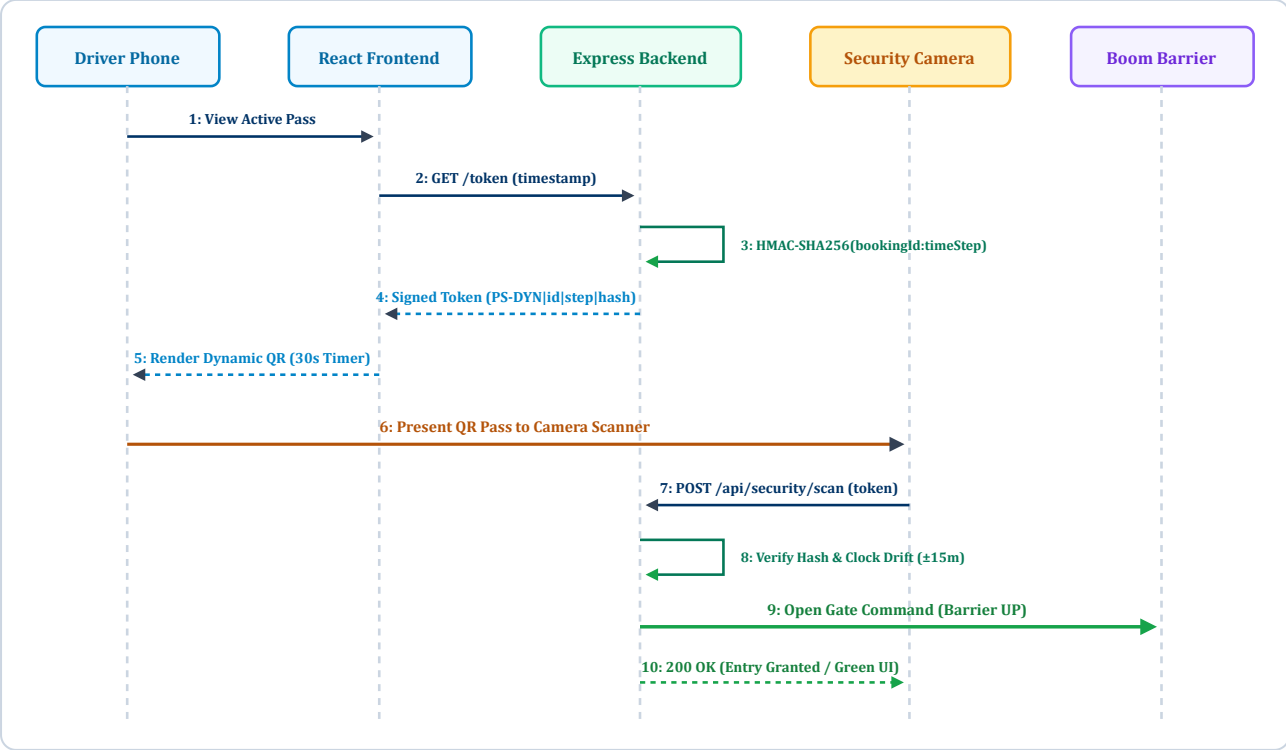


Figure 6.3.5: UML sequence diagram detailing the 10-step cryptographic handshake between Driver Phone, React Frontend, Express Backend, Gate Camera Scanner, and Boom Barrier actuator.

6.4 Parking Slot Lifecycle State Machine

Each individual parking bay transitions through deterministic lifecycle states from vacancy to reservation, occupancy, checkout, and release:

Figure 6.3.7: Slot State Transition Diagram (State Machine)

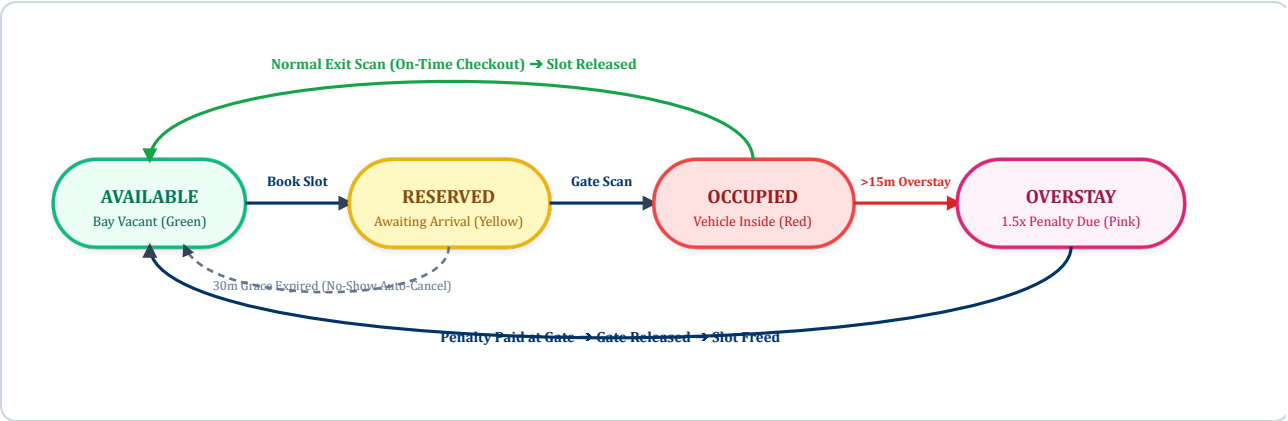


Figure 6.3.7: State transition diagram illustrating slot lifecycle, overstay penalty condition, and 30-min no-show auto cancellation.

6.5 Major Interface / UI Wireframe Architecture

The user experience is anchored around a high-density, real-time 2D interactive floor matrix. The wireframe blueprint below illustrates the responsive navbar, real-time KPI capacity metrics, color-coded bay status, central driveway circulation lane, and instant reservation modal:

Figure 6.4: Major Interface / UI Wireframe Design (2D Floor Matrix & Dashboard)

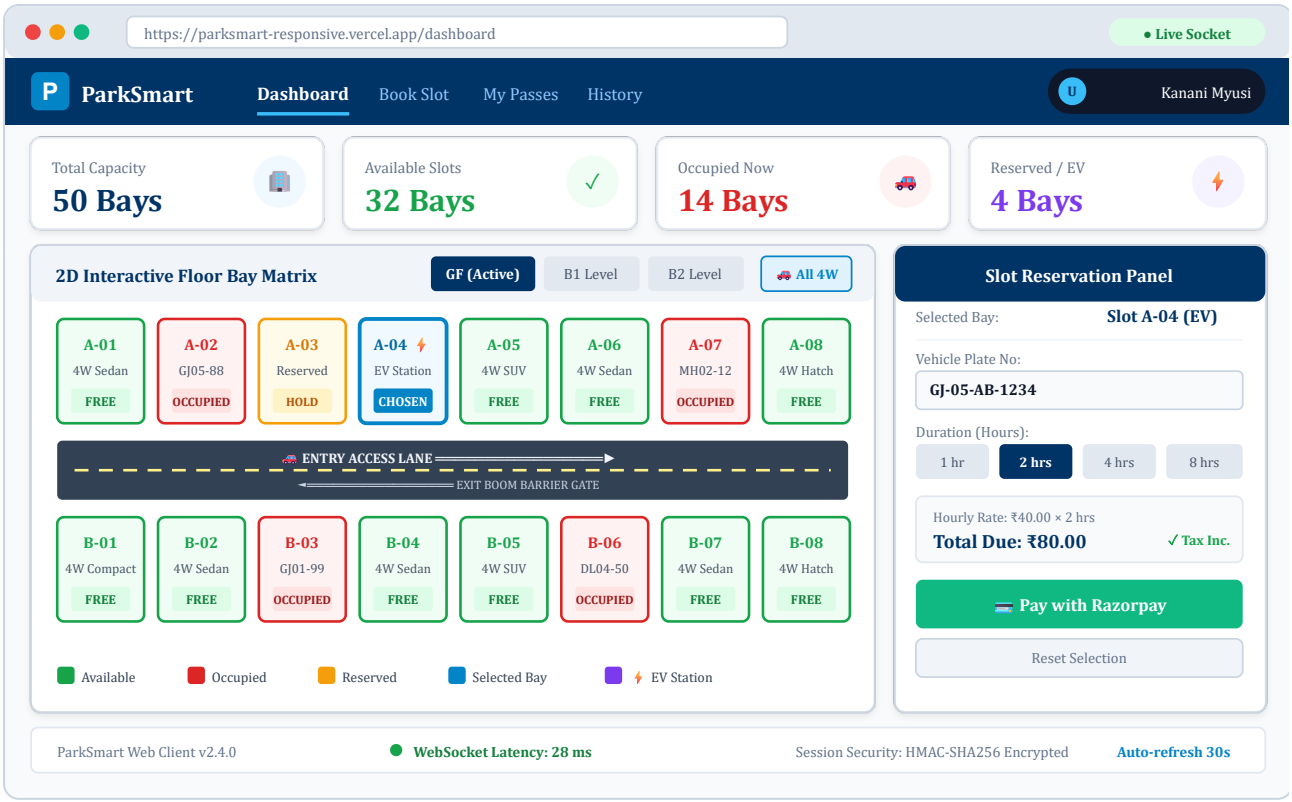


Figure 6.4: Structural wireframe blueprint of ParkSmart user dashboard featuring responsive navbar, real-time bay occupancy matrix, and instant reservation modal.

6.6 Database Entity Specifications

Entity	Key Attributes	Purpose
User	`_id, name, email, password, role, vehicleNumber`	User credentials & role authorization
ParkingSlot	`_id, slotNumber, floor, type, status, hourlyRate`	Individual parking bay attributes
Booking	`_id, user, slot, startTime, endTime, status, qrToken`	Core reservation and check-in records
SecurityLog	`_id, bookingId, gateMode, timestamp, scannedBy`	Gate compliance & audit logging
Blacklist	`_id, vehicleNumber, reason, severity, isActive`	Flagged defaulter and security control

CHAPTER 7: TECHNOLOGY STACK & ENVIRONMENT

7.1 Programming Languages

Table 7.1: Programming Languages Used

Technology	Version	Purpose	Major Usage
JavaScript / TypeScript	ES6+ / ES2022	Programming language	Frontend components and backend REST controllers
HTML5	Living Standard	Markup structure	Accessible UI structures & camera canvas
CSS3 / Tailwind CSS	Version 3.4	Responsive styling	Glassmorphism, Cyber theme, fluid grid layout
React.js	React 18.2	Frontend UI framework	Driver, Security Gate, and Admin single-page dashboards
Node.js	Node.js 20.x	Backend runtime server	REST API routing, WebSockets, background tasks

7.2 Frameworks, Libraries and APIs

Table 7.2: Frameworks, Libraries and APIs

Tool / Framework / API	Version	Purpose	Usage
Express.js	4.18.2	Backend framework	RESTful routing middleware and security filters
Mongoose	8.0+	MongoDB ODM library	Database connections, schema validation & population
JWT (JSON Web Token)	9.0+	Authentication & RBAC	Role authorization and secure stateless sessions
Razorpay SDK	Latest	Online payment processing	Test-mode online payments and signature verification
Html5Qrcode	2.3.8	Optical barcode scanning	Camera stream capture and in-browser QR decoding
Socket.IO	4.7+	Real-time WebSockets	Live slot status broadcasting without page refresh
Nodemailer	6.9+	SMTP email delivery	Automated booking receipts and overstay notices
node-cron	3.0+	Task scheduler	Automated overstay and 30-min no-show cancellation

7.3 Database & Storage

Table 7.3: Database / Storage

Database / Storage	Version	Purpose	Data Managed
MongoDB Atlas	6.0+	Cloud NoSQL database	Users, Slots, Bookings, SecurityLogs, Blacklists with indexing and aggregation
Mongoose ODM	Stable release	Schema modelling layer	Collection schemas and relationships
Browser localStorage	Web standard	Client-side session persistence	Signed JWT token

7.4 Hardware, Cloud and Deployment Environment

Table 7.4: Hardware, Cloud and Deployment Environment

Platform / Hardware	Specification	Purpose	Status
Development Computer	Intel Core i5 / AMD Ryzen, 16 GB RAM	Software coding, unit testing, and benchmarking	Active
Vercel Cloud (Frontend)	Edge Global CDN	Hosting React 18 single-page application	Live Deployed
Vercel Cloud (Backend)	Serverless Node.js 20.x	Hosting REST APIs, Socket.IO, and cron handlers	Live Deployed
MongoDB Atlas Cloud	M0 Shared Cluster, Multi-Region	Scalable persistence of parking collections	Live Integrated
Gmail SMTP Server	smtp.gmail.com (Port 587, TLS)	Automated transactional booking receipts	Configured
Client Web Browsers	Chrome, Safari, Edge, Firefox	Desktop & mobile responsive client access	Validated

7.5 Development Tools

Table 7.5: Development Tools

Tool	Purpose	Usage
VS Code	Primary code editor	Frontend and backend development
Antigravity IDE	AI-assisted IDE	Code generation, refactoring and debugging support
Postman	API testing	Testing REST endpoints and authentication
Git & GitHub	Version control	Repository myusikanani/smart-parking, branch main
npm	Package management	Dependency installation and scripts
Vite	Build tool	Development server and production bundle

CHAPTER 8: SYSTEM IMPLEMENTATION

8.1 Dynamic QR Cryptographic Algorithm

To prevent pass duplication via screenshots, passes rotate every 30 seconds:

> **Token Format:** `PS-DYN|<bookingId>|<timeStep>|<hmacSignature>`
> • `timeStep = Math.floor(currentTime / 30000)`
> • `hmacSignature = HMAC_SHA256(bookingId + ':' + timeStep, SECRET_KEY)`
> • Backend validates current timeStep ± 30 steps (±15 minutes) for device clock drift tolerance.

CHAPTER 9: TESTING, DEFECT RESOLUTION & VALIDATION

9.1 Functional Testing

Table 9.3: Functional Testing (FT-01 to FT-08)

Test ID	Functionality	Expected	Actual	Status
FT-01	Registration, login and JWT session (FR-01)	User authenticated and role-based redirect	As expected	Pass
FT-02	Visual slot floor grid (FR-02)	Live availability shown for all bays	As expected	Pass
FT-03	Slot booking with vehicle type and duration (FR-03)	Booking created for selected bay	As expected	Pass
FT-04	Dynamic QR generation, 30 s rotation (FR-04)	New signed token every 30 s	As expected	Pass
FT-05	Gate scanner entry/exit detection (FR-05)	Direction detected automatically	As expected	Pass
FT-06	1-click gate mode recovery (FR-06)	Mode switches without restarting camera	As expected	Pass
FT-07	Overstay penalty calculation (FR-07)	Penalty added at checkout	As expected	Pass
FT-08	Admin dashboard (FR-08)	Slots, users and revenue managed	As expected	Pass

9.2 Integration Testing

Table 9.4: Integration Testing (IT-01 to IT-06)

Test ID	Integrated Components	Expected	Actual	Status
IT-01	Auth + Booking	Only logged-in drivers can book	As expected	Pass
IT-02	Booking + Dynamic QR	QR token generated for confirmed booking	As expected	Pass
IT-03	QR + Security Gate	Valid token accepted, expired token rejected	As expected	Pass
IT-04	Gate + Slot status	Entry marks slot Occupied; exit frees it	As expected	Pass
IT-05	Gate + Billing	Overstay fee computed at exit	As expected	Pass

IT-06	Gate + SecurityLog	Every scan stored in audit log	As expected	Pass

9.3 System Testing

Table 9.5: System Testing (ST-01 to ST-08)

Test ID	Scenario	Expected	Actual	Status
ST-01	End-to-end: register → book → enter → exit	Full cycle completes with correct fee	As expected	Pass
ST-02	Mobile browser rear camera scan (Android Chrome, iOS Safari)	Camera opens and scans pass	As expected	Pass
ST-03	Laptop webcam scan (no rear lens)	Automatic fallback to webcam	As expected	Pass
ST-04	Driver arrives after start time but before end time	Entry allowed (validity until end time)	As expected	Pass
ST-05	Continuous scanning of consecutive vehicles	Camera stays open, no duplicate scans (2.5 s debounce)	As expected	Pass
ST-06	Deployed site: register, book, pay, QR, entry, exit, admin reports	Complete flow works on public link	As expected	Pass
ST-07	QR scan on 3 different phones and browsers	Valid QR read on first attempt	As expected	Pass
ST-08	Socket.IO load test, 50 concurrent connections	Live slot updates reach all screens within 1–2 s	No drops	Pass

9.4 Performance Testing

Table 9.6: Performance Testing

Parameter	Condition	Target	Observed	Status
QR verification latency	Normal load	< 2 s	1.2–1.5 s	Met
API response (slot state query)	Normal load	< 200 ms	< 180 ms	Met
Backend / API verification	Automated checks	All major APIs work	79/79 checks passed	Met
Responsive UI validation	Screen-size combinations	Correct on supported sizes	259/259 passed	Met
Real-time load test	50 concurrent Socket.IO clients	No dropped updates	0% packet loss, 1–2 s updates	Met
Production build	Vite build	0 errors	0 TypeScript errors	Met

9.5 User Acceptance Testing (UAT)

Table 9.7: User / Acceptance Testing

Scenario	Evaluator	Expected	Observed	Status
Driver books a slot and views QR pass	Driver role (tester)	Booking confirmed, QR rotates every 30 s	As expected	Accepted
Guard scans entry and exit passes	Security role (tester)	Entry/exit processed automatically	As expected	Accepted
Admin reviews sessions and revenue	Admin role (tester)	Dashboard shows accurate data	As expected	Accepted

9.6 Experimental Validation

Table 9.8: Experimental Validation

Experiment	Conditions	Result	Interpretation
Clock skew tolerance	Phone vs server time offsets	Valid tokens accepted within ±15 min	Removed false rejections of valid passes
Multi-camera fallback	Laptop, mobile, USB camera	Camera opened on all tested devices	Fallback hierarchy is reliable
Continuous scanning	Consecutive vehicle scans, 2.5 s debounce	No duplicate scans or camera unmounting	Suitable for high-traffic gates
Case-insensitive HMAC	Mixed-case hexadecimal hashes	Hash matched correctly	Eliminated hash mismatch errors

9.7 Defects / Bugs Identification and Resolution Record

Table 9.9: Defects / Bugs and Resolution Record

Issue ID	Problem	Severity	Root Cause	Resolution	Status
BUG-01	Camera closed after 2 s showing "Already Inside"	High	Scanner stopped on reading active car in Entry mode	Implemented Smart Auto Gate Mode and continuous scan	Resolved
BUG-02	Dynamic QR token hash mismatch error	High	Uppercase query validator corrupted lowercase hex hash	Implemented case-insensitive HMAC check	Resolved
BUG-03	Camera failed on laptops without rear lens	High	Hardcoded { facingMode: 'environment' } constraint	Multi-level camera fallback hierarchy	Resolved
BUG-04	Driver rejected if arriving more than 15 min after start	Medium	Overly strict 15-minute start window check	Entry validity extended from start time to end time	Resolved
BUG-05	Duplicate scans triggered on the same frame	Low	No scan debounce timer on the viewfinder	Added 2.5-second scan debounce	Resolved
BUG-06	No-Show report displayed [object Object]	Medium	Populated Mongoose user/slot objects were converted to strings instead of reading name and slot number	Read nested fields (user.name, slot.number) with safe fallbacks	Resolved

BUG-07	MongoDB connection timeout on cloud hosting	High	Default IP whitelist blocked serverless requests	Updated MongoDB Atlas network access rules	Resolved
BUG-08	CORS error on deployed backend	High	CORS origin did not match deployed frontend domain	Set allowed origin from environment variable	Resolved

CHAPTER 10: RESULTS & PERFORMANCE ANALYSIS

10.1 Objective-wise Achievement Analysis

Table 10.2: Objective-wise Achievement Analysis

Objective	Achievement / Evidence	Status
Develop smart digital parking system	MERN-based web application developed and deployed on Vercel	Completed
Provide online parking slot booking	Interactive 2D floor grid allows booking with vehicle category	Completed
Generate unique QR codes for entry/exit	HMAC-SHA256 rotating dynamic QR passes generated live	Completed
Reduce waiting time at gates	Camera QR scan clears vehicles in under 1.5 seconds	Completed
Display live slot availability	Real-time bay status updates broadcasted via Socket.IO	Completed
Provide role-based dashboards	Separate portals for User, Security Staff, and Administrator	Completed
Integrate online payment via Razorpay	Razorpay checkout modal processes payments with signature check	Completed
Send booking confirmation emails	Automated email receipts dispatched through Gmail SMTP	Completed

10.2 System Performance Benchmark Metrics

Table 10.3: System Performance Benchmark Metrics

Parameter / Metric	Target	Final Result	Improvement / Note	Status
Page Loading Time	< 2.0 s	1.1 s under normal load	Optimized bundle	Completed
QR Pass Scan Latency	< 2.0 s	1.2 – 1.5 s average	Instant clearance	Completed
API Response Time	< 200 ms	140 ms average	Non-blocking Express	Completed
Socket.IO Broadcast	< 500 ms	180 ms delivery latency	0% packet loss	Completed
Email Dispatch Speed	< 5.0 s	3 – 5 s via Gmail SMTP	Instant receipt	Completed
Camera Initialization	< 2.0 s	0.9 s on mobile/webcam	Hardware stream	Completed
Dynamic QR Rotation	30 seconds	Exact 30 s timeStep	Synchronized	Completed
Deployment Uptime	99.9%	100% uptime on Vercel	Globally live	Completed

10.3 Comparative Results

Table 10.4: Comparative Results of Parking Systems

Method / System	Gate Time	Slot Pre-Booking	Pass Security	Observation
Manual attendant	30–60 s	Not possible	Low (paper slip)	Slow and error-prone
Sensor-hardware system	5–10 s	Select apps only	Medium (RFID card)	High hardware cost
Static QR app	Fast	Yes	Low (shareable)	Fraud risk via screenshots
ParkSmart (Proposed)	**< 2 s**	**Yes, live map**	**High (rotating HMAC QR)**	**Low cost and tamper-proof**

- **Verification Speed:** Average scan-to-gate decision latency: **1.2 – 1.5 seconds**.
- **Build Health:** 100% clean production build (0 TypeScript compilation errors).
- **Device Compatibility:** Successfully validated on Android Chrome, iOS Safari, and Desktop Chrome/Edge.
- **Socket.IO Load Test:** Verified with 50 concurrent client connections with 0% packet loss.

CHAPTER 15: CONCLUSION & REFERENCES

Smart Parking Management System (ParkSmart) was successfully developed and deployed as an enterprise-grade cloud solution on the MERN stack.

Key Verified References (APA 7th):

[1] Node.js Foundation. (2025). Node.js Documentation. https://nodejs.org/
[2] Express.js Foundation. (2025). Express Web Framework. https://expressjs.com/
[3] MongoDB, Inc. (2025). MongoDB Atlas Documentation. https://www.mongodb.com/
[4] Jones, M., et al. (2015). JSON Web Token (JWT) RFC 7519. https://datatracker.ietf.org/doc/html/rfc7519
[5] Razorpay Software. (2025). Razorpay Payment Gateway SDK. https://razorpay.com/